

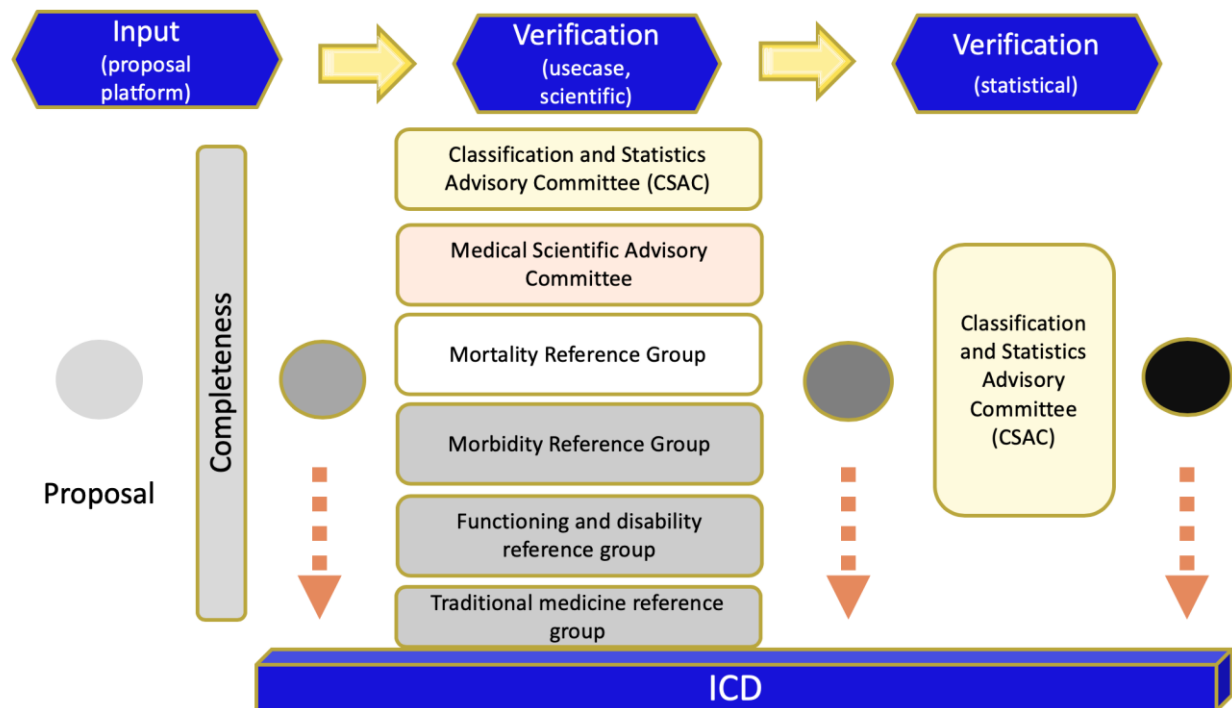
### 3.12.5 Proposal Workflow

ICD maintenance is a process that requires broad expertise in statistics and medical science as well as in different use cases. It has already been established that the structure associated with ICD update and maintenance will be revised in line with new requirements and efficient use of limited resources. The figure below shows the flow by which proposals received through the update platform might move through the expert groups for review and recommendation to WHO. A 'proposal triage' will initially review all proposals received to ensure that they are complete and have been submitted correctly. The proposals will then be forwarded to the appropriate next step(s). This triage may identify proposals that can or must be processed in an accelerated process, such as due to the type of proposal or in cases of extreme urgency. The proposed typology of proposals may help to identify the impact on the classification which will, in turn, inform whether the proposal is an improvement or clarification that can be implemented on an ongoing basis, or if the change must be queued for implementation on the updating cycle. The threshold for considering proposals to the foundation is low, allowing for simple rules to identify what is 'receivable/actionable' namely, if it is complete and correct, it may be considered. Consideration does not guarantee acceptance.

The CSAC is composed of:

- WHO-FIC Collaborating centres and officially nominated technical representatives from WHO member states. The participants and observers have specific knowledge and expertise in the classifications
- Two cochairs from each Reference Group (Mortality Reference Group (MRG), Morbidity Reference Group (MbRG) and Functioning and Disability Reference Group (FDRG), Traditional Medicine Group (TMRG)) as well as two cochairs from the Medical and Scientific Advisory Committee (MSAC) are also included ex officio, with the right to comment but no right to vote.
- Additional experts may be invited to participate in CSAC meetings for commenting on specific items.

All decisions are based on consensus. Terms of reference of the working groups are included in the document that describes the [conduct of the WHO Family of International Classifications Network](#).



### Proposal Workflow

Proposal Workflow see [Conduct of the WHOFIC Network](#) for ToR and Governance of the different groups.

### CSAC Small Group

This group consists of the CSAC co-chairs and secretariats, WHO staff and experts selected by WHO.

The CSAC Small Group reviews each proposal and determines if the proposal is ready for review by the full CSAC. The Small Group also makes a recommendation by consensus as to whether the proposal should be accepted or rejected. If a proposal requires more information or is incomplete, the CSAC Small Group may request further advice from MSAC or other committees and reference groups (C&RG) or send the proposal back to the author. Sometimes, the Small Group, MSAC and/or a C&RG make a recommendation for a modification to the original proposal, and this is recorded in the public comments section of the proposal by the secretariat of the group. These comments and proposed modifications are then considered by the full CSAC.

When a proposal is ready for review by the wider CSAC, the CSAC Secretariat records the recommendation of the CSAC Small Group, and any suggested modifications and the reasons for these, on the public comments section of the proposal and allocates the proposal to have the status 'CSAC Voting'. CSAC members can then indicate their agreement and/or comment on the proposal, as modified by the CSAC Small Group, as applicable.

### 3.12.6 Changes that cannot be done during the normal updating process

Changes that create new structures that conflict with the existing structure or coding of a current revision can be carried out only within a new revision of ICD and they include:

1. Changing an existing code of a category
2. Changing an existing grouping into a category if it has children in the linearisation (Allowing this would force changing the codes of the children therefore it is not allowed)
3. Changing an existing category into a grouping if it has children in the linearisation (Same as above)
4. Inserting a category in between two existing sibling categories. Categories to an existing siblinghood needs to be entered at the end before the residuals.
5. Re-ordering of any kind will not be possible as the ordering would be based on the codes which will not change.
6. Retiring an entity that contains children that are not retired and replacing it with another entity that contains the old children. (This is not possible because the new entity cannot keep the same code of the retired one and if we give a new code its code will not be compatible with the children.)
7. Moving an existing category in a way that would imply changing its code. E.g. the category 2B01.2 cannot be moved which is under [2B01](#) as a child of 2B00

For Chapter X 'Extension codes' the only limitation is number 1 above. Any other change could be done as the codes of Chapter X are not based on their hierarchical location.

### 3.12.7 Applicability and Intellectual Property

The following paragraph provides an extract of some of the legal regulations in relation to ICD. It is noted that this text does not prejudice in any way the wording of any legal arrangements that are made between WHO and other relevant parties. The binding official version of the ICD license is available at <https://icd.who.int/en>.

The ICD is the intellectual property of WHO and changes to the ICD are subject to contractual arrangements and approval through the updating mechanism.

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- - b. Reproduce and distribute ICD-11 in part or whole without the ICD-11 codes;
- - c. Reproduce ICD-11 in part or a whole without the ICD-11 URIs; or
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The ICD may be translated into any language. For translation, interested parties (the Translator) are requested to contact the WHO and comply with the relevant regulations and receive written permission in a formal contract. Usually translation rights are granted only to ministries of health.

The translations into the UN languages are a product of the WHO and all rights are vested with WHO. Translations into other languages are a product of the Translator.

The Translator will use the WHO translation platform, thus allowing the WHO to verify correctness and completeness of the translation.

WHO is automatically granted a perpetual and irrevocable, non-exclusive, world-wide, royalty-free, sub-licensable right to use, change, adapt, translate, publish, and disseminate such work product in any manner and in any format in conjunction with the work of WHO.

In some instances, users determine the need to change parts of the ICD in order to produce a special version of ICD. Production of special versions requires a dedicated contractual arrangement with WHO.

Such versions will be produced from the WHO production platform by WHO.

All changes or suggestions for changes to the ICD must be submitted to the WHO-ICD maintenance platform (for details see [3.12 Annex A: ICD-11 Updating and Maintenance](#)).

Requests for production of a special version will be subject to the provision of resources necessary for the WHO to generate the specialty version.

No publicity or advertising may be displayed in the coding or browsing pages. In case of embedding of the classification in a local website, or running a local version, a link to the ICD homepage at the WHO must be included.

No publicity or advertising may be displayed in the ICD print versions.

Ideally users will access the ICD online or through the web services provided by the WHO. This will ensure proper joint use of the index, content model, and tabular lists and facilitate dissemination of updates, where applicable. Any coding mechanism produced by 3rd parties must provide the same coding results as the reference online coding tool.

For national and international recording and reporting, the most up to date version of the ICD is to be used, as stipulated in the Nomenclature Regulations (1967).

### **Modifications for morbidity coding**

The use of ICD in the specific context of the health care system of a country may require detail that is not currently part of ICD-11, for example, due to specific settings or due to reimbursement system requirements. Such changes will be subject to the same international process as are all other changes to ICD and will then become part of the Foundation Component and eventually of the Mortality and Morbidity Statistics (MMS), ideally prior to their implementation in the requesting country.

A situation may arise, where a national government or a related national institution needs a modification to be implemented immediately. In such circumstances, conflicts with the current Foundation Component must be avoided, and the relevant changes will be subject to special mechanisms for the international updating process. All countries planning to produce modifications must make relevant contractual arrangements with WHO. This includes regulations on distribution within and outside the respective country and the resources necessary for the WHO to add such changes to the Foundation.

For developing a national modification of ICD-11 the following rules must be adhered to: 1. Modifications will be agreed by the ICD-11 maintenance bodies before they are implemented nationally. 2. Modifications must not impact on morbidity and mortality statistics, and must not conflict with the Foundation. 3. Approval of all modifications will be subject to consideration of whether suitable additional detail already exists in the Foundation. 4. If a change is made to the international version of ICD-11 the respective modification must incorporate the change as soon as possible.

### **Example**

'Diabetes Type 1' in the WHO version of ICD-11 is classified to [5A10](#). A national modification may require additional detail to be added to the ICD-11 codes. For example, 'Diabetes Type 1, uncontrolled' could be added as a subcategory to [5A10](#), as 5A10.1 Diabetes Type 1, uncontrolled. However, when the mechanism of postcoordination provides the necessary detail, the addition of a new subcategory may not be necessary.

### 3.13 Annex B: History of the development of the ICD

#### 3.13.1 Early history

Sir George Knibbs, the eminent Australian statistician, credited François Bossier de Lacroix (1706-1777), better known as Sauvages, with the first attempt to classify diseases systematically (1). Sauvages' comprehensive treatise was published under the title *Nosologia methodica*. A contemporary of Sauvages was the great methodologist Linnaeus (1707-1778), author of *Genera morborum*, a catalogue of diseases. More recently, Moriyama et al (2) have referred to 16th century and 17th predecessors Fernel and Sydenham. At the beginning of the 19th century, the classification of disease in most general use was one by William Cullen (1710- 1790, of Edinburgh, which was published in 1785 under the title *Synopsis nosologiae methodicae*.

For all practical purposes, however, the statistical study of disease began a century earlier with the work of John Graunt on the London Bills of Mortality published in 1662. The kind of classification envisaged by this pioneer is exemplified by his attempt to estimate the proportion of liveborn children who died before reaching the age of six, when no records of age at death were available. He took all deaths and classed them using terms from the time, such as thrush, convulsions, rickets, teeth and worms, abortives, chrysomes, infants, and livergrown. Overlaid with these, he added to them deaths classed as smallpox, swinepox, measles, and worms without convulsions. Despite the crudity of this classification, his estimate of 36% mortality before the age of six years appears from later evidence to have been a good one. While three centuries have contributed to the scientific accuracy of disease classification, there are many who doubt the usefulness of attempts to compile statistics of disease, or even causes of death, because of the difficulties of classification. To these, one can quote Major Greenwood: '*The scientific purist, who will wait for medical statistics until they are nosologically exact, is no wiser than Horace's rustic waiting for the river to flow away*' (3).

Fortunately for the progress of preventive medicine, the General Register Office of England and Wales, at its inception in 1837, found in William Farr (1807-1883) – its first medical statistician. Farr was a man who not only made the best possible use of the imperfect classifications of disease available at the time but laboured to secure better classifications and international uniformity in their use.

Farr found Cullen's classification in use in the public services. It had not been revised to embody the advances in medical science, nor was it deemed by him to be satisfactory for statistical purposes. Farr realised that small numbers that would result from a detailed classification would not permit statistical inferences to be made. In the first Annual Report of the Registrar General (4), therefore, he discussed the principles that should govern a statistical classification of disease and urged the adoption of a uniform classification as follows:

\_The advantages of a uniform statistical nomenclature, however imperfect, are so obvious, that it is surprising no attention has been paid to its enforcement in Bills of Mortality. Each disease has, in many instances, been denoted by three or four terms, and each term has been applied to as many different diseases: vague, inconvenient names have been employed, or complications have been registered ins

stead of primary diseases. The nomenclature is of as much importance in this department of inquiry as weights and measures in the physical sciences, and should be settled without delay.\_

Both nomenclature and statistical classification received constant study and consideration by Farr in his annual 'Letters' to the Registrar General published in the Annual Reports of the Registrar General. Farr did much to promote his classification but could not find general acceptance (4). However, the utility of a uniform classification of causes of death was so strongly recognised at the first International Statistical Congress, held in Brussels in 1853, that the Congress requested William Farr and Genevan Marc d'Espine to prepare an internationally applicable, uniform classification of causes of death.

At the next Congress, in Paris in 1855, Farr and d'Espine submitted two separate lists which were based on very different principles. Farr's classification was arranged under five groups: epidemic diseases, constitutional (general) diseases, local diseases arranged according to anatomical site, developmental diseases, and diseases that are the direct result of violence. D'Espine classified diseases according to their nature (gouty, herpetic, haematic, etc.). The Congress adopted a compromise list of 139 rubrics. In 1864, this classification was revised in Paris on the basis of Farr's model and was subsequently further revised in 1874, 1880, and 1886. Although this classification was never universally accepted, the general arrangement proposed by Farr, including the principle of classifying diseases by aetiology followed by anatomical site, survived as the basis of the International List of Causes of Death.

Importantly, the 1855 Congress also recommended that each country should ask for information on causes of death from the doctor who had been attending the deceased, and that each country should take measures to ensure that all deaths were verified by doctors (4).

### 3.13.2 Adoption of the International List of Causes of Death

At its 1891 meeting in Vienna, the International Statistical Institute, the successor to the International Statistical Congress, charged a committee chaired by Jacques Bertillon (1851-1922), Chief of Statistical Services of the City of Paris, with the preparation of a classification of causes of death. The committee's report was presented and adopted at the meeting of the International Statistical Institute in Chicago in 1893.

For main headings, Bertillon adopted the anatomical site rather than the nature of disease, according to Farr's plan. Bertillon's list included defined diseases most worthy of study by reason of their transmissible nature or their frequency of occurrence. In accordance with the instructions of the Vienna Congress, Bertillon included three classifications: an abridged classification of 44 titles; a classification of 99 titles; and a classification of 161 titles. Bertillon also prepared some rules or guidelines on the resolution of problems; for example, how statistical clerks should classify what is written without imputing what the doctor might have meant (5). The 'Bertillon Classification of Causes of Death', as it was first called, received general approval and was adopted by several countries, as well as by many cities. The classification was first used in North America by Jesus E. Monjaras for the statistics of San Luis de Potosi, Mexico (5). In 1898, the American Public Health Association, at its meeting in Ottawa, Canada, recommended the adoption of the Bertillon Classification by

registrars of Canada, Mexico, and the United States of America. The Association further suggested that the classification should be revised every ten years.

At the meeting of the International Statistical Institute at Christiania in 1899, Bertillon presented a report on the progress of the classification, including the recommendations of the American Public Health Association for decennial revisions. The International Statistical Institute then adopted the following resolution (6): The International Statistical Institute, convinced of the necessity of using comparable nomenclatures in the different countries:

\_Learns with pleasure of the adoption by all the statistical offices of North America, by some of those of South America, and by some in Europe, of the system of cause of death nomenclature presented in 1893; Insists vigorously that this system of nomenclature be adopted in principle and without revision, by all the statistical institutions of Europe; Approves, at least in its general lines, the system of decennial revision proposed by the American Public Health Association at its Ottawa session (1898); Urges the statistical offices who have not yet adhered, to do so without delay, and to contribute to the comparability of the cause of death nomenclature.\_

The French Government therefore assembled in Paris, in August 1900, the first International Conference for the Revision of the Bertillon or International List of Causes of Death. Delegates from 26 countries attended this Conference. A detailed classification of causes of death consisting of 179 groups and an abridged classification of 35 groups was adopted on 21 August 1900. The desire for decennial revisions was recognized, and the French Government was requested to call the next meeting in 1910. In fact, the next conference was held in 1909, and the Government of France called succeeding conferences in 1920, 1929, and 1938. Bertillon continued to be the guiding force in the promotion of the International List of Causes of Death, and the revisions of 1900, 1910, and 1920 were carried out under his leadership. As Secretary-General of the International Conference, he sent out the provisional revision for 1920 to more than 500 people, asking for comments. His death in 1922 left the International Conference without a guiding hand.

At the 1923 session of the International Statistical Institute, Michel Huber, Bertillon's successor in France, recognized this lack of leadership and introduced a resolution for the International Statistical Institute to renew its stand of 1893 in regard to the International Classification of Causes of Death and to cooperate with other international organisations in preparation for subsequent revisions. The Health organisation of the League of Nations had also taken an active interest in vital statistics and appointed a Commission of Statistical Experts to study the classification of diseases and causes of death, as well as other problems in the field of medical statistics. E. Roesle, Chief of the Medical Statistical Service of the German Health Bureau and a member of the Commission of Statistical Experts, prepared a monograph that listed the expansion in the rubrics of the 1920 International List of Causes of Death that would be required if the classification were to be used in the tabulation of statistics of morbidity. This careful study was published by the Health organisation of the League of Nations in 1928 (7). In order to coordinate the work of both agencies, an international 'Mixed Commission' was created with an equal number of representatives from the International Statistical Institute and the Health organisation of the League of



Nations. This Commission drafted the proposals for the Fourth (1929) and the Fifth (1938) revisions of the International List of Causes of Death.

### 3.13.3 The Fifth Decennial Revision Conference

The Fifth International Conference for the Revision of the International List of Causes of Death, like the preceding conferences, was convened by the Government of France and was held in Paris in October 1938. The Conference approved three lists: a detailed list of 200 titles, an intermediate list of 87 titles and an abridged list of 44 titles. Apart from bringing the lists up to date in accordance with the progress of science, particularly in the chapter on infectious and parasitic diseases, and changes in the chapters on puerperal conditions and on accidents, the Conference made as few changes as possible in the contents, number, and even in the numbering of the items. A list of causes of stillbirth was also drawn up and approved by the Conference.

As regards classification of diseases for morbidity statistics, the Conference recognised the growing need for a corresponding list of diseases to meet the statistical requirements of widely differing organisations, such as health insurance organisations, hospitals, military medical services, health administrations, and similar bodies. The following resolution, therefore, was adopted (8):

#### 3.13.3.1 International Lists of Diseases

- In view of the importance of the compilation of international lists of diseases corresponding to the international lists of causes of death: The Conference recommends that the Joint Committee appointed by the International Institute of Statistics and the Health organisation of the League of Nations undertake, as in 1929, the preparation of international lists of diseases, in conjunction with experts and representatives of the organisations specially concerned. Pending the compilation of international lists of diseases, the Conference recommends that the various national lists in use should, as far as possible, be brought into line with the detailed International List of Causes of Death (the numbers of the chapters, headings and subheadings in the said List being given in brackets). The Conference further recommended that the United States Government continue its studies of the statistical treatment of joint causes of death in the following resolution (9):
  - Death Certificate and Selection of Causes of Death where more than One Cause is given (Joint Causes) The Conference,
    - Whereas, in 1929, the United States Government was good enough to undertake the study of the means of unifying the methods of selection of the main cause of death to be tabulated in those cases where two or more causes are mentioned on the death certificate,
    - And whereas, the numerous surveys completed or in the course of preparation in several countries reveal the importance of this problem, which has not yet been solved,
    - And whereas, according to these surveys, the international comparability of death rates from the various diseases requires, not only the solution of the problem of the selection of the main

tabulated cause of death, but also the solution of several other questions;

- Warmly thanks the United States Government for the work it has accomplished or promoted in this connection;
- Requests the United States Government to continue its investigations during the next ten years, in cooperation with other countries and organisations, on a slightly wider basis, and
- Suggests that, for these future investigations, the United States Government should set up a subcommittee comprising representatives of countries and organisations participating in the investigations undertaken in this connection.

#### 3.13.4 Previous classifications of diseases for morbidity statistics

In the discussion so far, classification of disease has been presented almost wholly in relation to cause-of-death statistics. Farr, however, recognized that it was desirable “to extend the same system of nomenclature to diseases which, though not fatal, cause disability in the population, and now figure in the tables of the diseases of armies, navies, hospitals, prisons, lunatic asylums, public institutions of every kind, and sickness societies, as well as in the census of countries like Ireland, where the diseases of all the people are enumerated” (10). In his *‘Report on nomenclature and statistical classification of diseases’*, presented to the Second International Statistical Congress, he therefore included in the general list of diseases most of those diseases that affect health as well as diseases that are fatal. At the Fourth International Statistical Congress, held in London in 1860, Florence Nightingale urged the adoption of Farr’s classification of diseases for the tabulation of hospital morbidity in the paper, *‘Proposals for a uniform plan of hospital statistics’*.

At the First International Conference to revise the Bertillon Classification of Causes of Death in Paris in 1900, a parallel classification of diseases for use in statistics of sickness was adopted. A parallel list was also adopted at the second conference in 1909. The extra categories for non-fatal diseases were formed by subdivision of certain rubrics of the cause-of-death classification into two or three disease groups, each of these being designated by a letter. The translation in English of the Second Decennial Revision, published by the United States Department of Commerce and Labor in 1910, was entitled International Classification of Causes of Sickness and Death. Later revisions incorporated some of the groups into the detailed International List of Causes of Death. The Fourth International Conference adopted a classification of illness which differed from the detailed International List of Causes of Death only by the addition of further subdivisions of 12 titles. These international classifications of illnesses, however, failed to receive general acceptance, as they provided only a limited expansion of the basic cause-of-death list.

In the absence of a uniform classification of diseases that could be used satisfactorily for statistics of illness, many countries found it necessary to prepare their own lists. A Standard Morbidity Code was prepared by the Dominion Council of Health of Canada and published in 1936. The main subdivisions of this code represented the 18 chapters of the 1929 Revision of the International List of Causes of Death, and these were subdivided into some 380 specific disease categories. At the Fifth International Conference in 1938, the Canadian delegate introduced a modification of this list for consideration as the basis for an international list of causes of illness. Although no action was taken on this proposal, the Conference adopted the resolution quoted above.

In 1944, provisional classifications of diseases and injuries were published in both the United Kingdom and the United States for use in the tabulation of morbidity statistics. Both classifications were more extensive than the Canadian list, but, like it, followed the general order of diseases in the International List of Causes of Death. The British classification was prepared by the Committee on Hospital Morbidity Statistics of the Medical Research Council, which was created in January 1942. It is entitled '*A provisional classification of diseases and injuries for use in compiling morbidity statistics*' (8). It was prepared with the purpose of providing a scheme for collecting and recording statistics of patients admitted to hospitals in the United Kingdom, using a standard classification of diseases and injuries, and was used throughout the country by governmental and other agencies.

A few years earlier, in August 1940, the Surgeon-General of the United States Public Health Service and the Director of the United States Bureau of the Census published a list of diseases and injuries for tabulation of morbidity statistics (9). The code was prepared by the Division of Public Health Methods of the Public Health Service in cooperation with a committee of consultants appointed by the Surgeon-General. '*The Manual for coding causes of illness according to a diagnosis code for tabulating morbidity statistics*', consisting of the diagnosis code, a tabular list of inclusions, and an alphabetical index, was published in 1944. The code was used in several hospitals, in a large number of voluntary hospital insurance plans and medical care plans, and in special studies by other agencies in the United States.

### 3.13.5 United States Committee on Joint Causes of Death

In compliance with the resolution of the Fifth International Conference, the American Secretary of State in 1945 appointed the United States Committee on Joint Causes of Death under the chairmanship of Lowell J. Reed, Professor of Biostatistics at Johns Hopkins University. Members and consultants of this committee included representatives of the Governments of Canada and the United Kingdom and the Health Section of the League of Nations. Recognizing a trend, the committee decided that it would be advantageous to consider classifications from the point of view of morbidity and mortality, since the problem of joint causes pertained to both types of statistics.

The committee also took into account that part of the resolution on International Lists of Diseases of the previous International Conference recommending that the 'various national lists in use should, as far as possible, be brought into line with the detailed International List of Causes of Death'. It recognized that the classification of sickness and injury is closely linked with the classification of causes of death. The view that such lists are fundamentally different arises from the erroneous belief that the International List is a classification of terminal causes, whereas it is in fact based upon the morbid condition that initiated the train of events ultimately resulting in death. The committee believed that, in order to utilise fully both morbidity and mortality statistics, not only should the classification of diseases for both purposes be comparable, but if possible, there should be a single list.

Furthermore, an increasing number of statistical organisations were using medical records involving both sickness and death. Even in organisations that compile only morbidity statistics, fatal as well as non-fatal cases needed to be coded. A single list, therefore, greatly facilitates their coding operations. It also provides a common base for comparison of morbidity and mortality statistics.

A subcommittee was therefore appointed, which prepared a draft of a Proposed Statistical Classification of Diseases, Injuries and Causes of Death. A final draft was adopted by the committee after it had been modified on the basis of trials undertaken by various agencies in Canada, the United Kingdom and the United States of America.

**Table 1:** *ICD Revisions under the auspices of WHO*

| <b>Revision</b> | <b>Date of coming into effect</b> | <b>Year of WHA adoption</b> |
|-----------------|-----------------------------------|-----------------------------|
| 6th Revision    | 1948                              | 1948 (WHA1.36)              |
| 7th Revision    | 1 Jan 1958                        | May 1956 (WHA9.29)          |
| 8th Revision    | 1 Jan 1968                        | May 1966 (WHA19.44)         |
| 9th Revision    | 1 Jan 1979                        | May 1976 (WHA29.34)         |
| 10th Revision   | 1 Jan 1993                        | May 1990 (WHA43.24)         |
| 11th Revision   | 1 Jan 2022                        | May 2019 (WHA72.15)         |

### 3.13.6 Sixth Revision of the International Lists

The International Health Conference held in New York City in June and July 1946 (11) entrusted the Interim Commission of the World Health Organisation with the responsibility of 'reviewing the existing machinery and of undertaking such preparatory work as may be necessary in connection with: (i) the next decennial revision of 'The International Lists of Causes of Death' (including the lists adopted under the International Agreement of 1934, relating to Statistics of Causes of Death); and (ii) the establishment of International Lists of Causes of Morbidity.'

To meet this responsibility, the Interim Commission appointed the Expert Committee for the Preparation of the Sixth Decennial Revision of the International Lists of Diseases and Causes of Death. This Committee, taking full account of prevailing opinion concerning morbidity and mortality classification, reviewed and revised the above-mentioned proposed classification which had been prepared by the United States Committee on Joint Causes of Death.

The resulting classification was circulated to national governments preparing morbidity and mortality statistics for comments and suggestions under the title, International Classification of Diseases, Injuries, and Causes of Death. The Expert Committee considered the replies and prepared a revised version incorporating changes to improve the utility and acceptability of the classification. The Committee also compiled a list of diagnostic terms to appear under each title of the classification. Furthermore, a subcommittee was appointed to prepare a comprehensive alphabetical index of diagnostic statements classified to the appropriate category of the classification. The Committee also considered the structure and uses of special lists of causes for tabulation and publication of morbidity and mortality statistics and studied other problems related to the international comparability of mortality statistics, such as form of medical certificate and rules for classification. The International Conference for the Sixth Revision of the International Lists of Diseases and Causes of Death was convened in Paris from 26 to 30 April 1948 by the Government of France under the terms of the agreement signed at the close of the Fifth Revision Conference in 1938. Its secretariat was entrusted jointly to the competent French authorities and to the World Health Organisation, which had carried out the preparatory work under the terms of the

arrangement concluded by the governments represented at the International Health Conference in 1946 (12).

The Conference adopted the classification prepared by the Expert Committee as the Sixth Revision of the International Lists (13). It also considered other proposals of the Expert Committee concerning the compilation, tabulation and publication of morbidity and mortality statistics. The Conference approved the International Form of Medical Certificate of Cause of Death, accepted the underlying cause of death as the main cause to be tabulated, and endorsed the rules for selecting the underlying cause of death as well as the special lists for tabulation of morbidity and mortality data. It further recommended that the World Health Assembly should adopt regulations under Article 21(b) of the WHO Constitution to guide Member States in compiling morbidity and mortality statistics in accordance with the International Statistical Classification. In 1948, the First World Health Assembly endorsed the report of the Sixth Revision Conference and adopted World Health Organisation Regulations No. 1, devised based on the recommendations of the Conference. The International Classification, including the Tabular List of Inclusions defining the content of the categories, was incorporated, together with the form of the medical certificate of cause of death, the rules for classification and the special lists for tabulation, into the Manual of the International Statistical Classification of Diseases, Injuries, and Causes of Death (22). The Manual consisted of two volumes, Volume 2 being an alphabetical index of diagnostic terms coded to the appropriate categories. In the Sixth Revision, morbid conditions resulting from injuries, poisonings and other external causes were classified according to both the external circumstances giving rise to the injury and to the kind of injury.

The adoption of this dual classification was regarded at the time as a bold step to deal with the simultaneous interest in more than one aspect of injury. The Sixth Decennial Revision Conference marked the beginning of a new era in international vital and health statistics. Apart from approving a comprehensive list for both mortality and morbidity and agreeing on international rules for selecting the underlying cause of death, it recommended the adoption of a comprehensive programme of international cooperation in the field of vital and health statistics. An important item in this programme was the recommendation that governments establish national committees on vital and health statistics to coordinate the statistical activities in the country, and to serve as a link between the national statistical institutions and the World Health Organisation. It was further envisaged that such national committees would, either singly or in cooperation with other national committees, study statistical problems of public health importance and make the results of their investigations available to the WHO.

### 3.13.7 The Seventh and Eighth Revisions

The International Conference for the Seventh Revision of the International Classification of Diseases was held in Paris under the auspices of the WHO in February 1955 (14). In accordance with a recommendation of the WHO Expert Committee on Health Statistics, this revision was limited to essential changes and amendments of errors and inconsistencies (15). The Eighth Revision Conference convened by the WHO met in Geneva, from 6 to 12 July 1965 (16). This revision was more radical than the Seventh but left unchanged the basic structure of the Classification and the general philosophy of classifying diseases, whenever possible, according to their aetiology rather than a particular manifestation. During the

years that the Seventh and Eighth Revisions of the ICD were in force, the use of the ICD for indexing hospital medical records increased rapidly and some countries prepared national adaptations which provided the additional detail needed for this application of the ICD.

### 3.13.8 The Ninth Revision

The International Conference for the Ninth Revision of the International Classification of Diseases, convened by the WHO, met in Geneva from 30 September to 6 October 1975 (17). In the discussions leading up to the conference, it had originally been intended that there should be little change other than updating of the classification. This was mainly because of the expense of adapting data processing systems each time the classification was revised. There had been an enormous growth of interest in the ICD and ways had to be found of responding to this, partly by modifying the classification itself and partly by introducing special coding provisions.

A number of representations were made by specialist bodies which had become interested in using the ICD for their own statistics. Some subject areas in the classification were regarded as inappropriately arranged and there was considerable pressure for more detail and for adaptation of the classification to make it more relevant for the evaluation of medical care, by classifying conditions to the chapters concerned with the part of the body affected rather than to those dealing with the underlying generalised disease. At the other end of the scale, there were representations from countries and areas where a detailed and sophisticated classification was irrelevant, but which nevertheless needed a classification based on the ICD in order to assess their progress in health care and in the control of disease. The final proposals presented to and accepted by the Conference retained the basic structure of the ICD, although with much additional detail at the level of the four-digit subcategories, and some optional five-digit subdivisions. For the benefit of users not requiring such detail, care was taken to ensure that the categories at the three-digit level were appropriate.

For the benefit of users wishing to produce statistics and indexes oriented towards medical care, the Ninth Revision included an optional alternative method of classifying diagnostic statements, including information about both an underlying general disease and a manifestation in a particular organ or site. This system became known as the dagger and asterisk system. The Twenty Ninth World Health Assembly, noting the recommendations of the International Conference for the Ninth Revision of the International Classification of Diseases, approved the publication, for trial purposes, of supplementary classifications of Impairments and Handicaps and of Procedures in Medicine as supplements to, but not as integral parts of, the International Classification of Diseases.

### 3.13.9 The Tenth Revision

Even before the Conference for the Ninth Revision, the WHO had been preparing for the Tenth Revision. It recognised that the great expansion in the use of the ICD necessitated a thorough rethinking of its structure and an effort to devise a stable and flexible classification, which should not require fundamental revision for many years to come. The WHO Collaborating Centres for Classification of Diseases (see [www.who.int/classification](http://www.who.int/classification)) were consequently called upon to experiment with models of alternative structures for ICD–10.

It had also become clear that the established ten-year interval between revisions was too short. Work on the revision process had to start before the current version of the ICD had been in use long enough to be thoroughly evaluated, mainly because the necessity to consult so many countries and organisations made the process a very lengthy one. The Director General of the WHO therefore wrote to the Member States and obtained their agreement to postpone a 1985 Tenth Revision Conference until 1989, and to delay the introduction of the Tenth Revision which would have been due in 1989. In addition to permitting experimentation with alternative models for the structure of the ICD, this allowed time for the evaluation of ICD-9, for example through meetings organised by some of the WHO Regional Offices and through a survey organised at headquarters.

The International Conference for the Tenth Revision of the International Classification of Diseases, attended by delegates from 43 Member States, was convened by the World Health organisation in Geneva from 26 September to 2 October 1989. The United Nations, the International Labour Organisation, and the WHO Regional Offices sent representatives to participate in the Conference, as did the Council for International organisations of Medical Sciences. Twelve other non-governmental organisations concerned with cancer registration, the deaf, epidemiology, family medicine, gynaecology and obstetrics, hypertension, health records, preventive and social medicine, neurology, psychiatry, rehabilitation and sexually transmitted diseases were also invited.

Extensive preparatory activity had been devoted to a radical review of the suitability of the structure of the ICD, essentially a statistical classification of diseases and other health problems, to serve a wide variety of needs for mortality and health care data. Ways of stabilizing the coding system to minimize disruption at successive revisions had been investigated, as had the possibility of providing a better balance between the content of the different chapters of the ICD. Even with a new structure, it was plain that one classification could not cope with the extremes of the requirements. The concept had therefore been developed of a 'family' of classifications, which would include the ICD for traditional mortality and morbidity statistics, while needs for more detailed, less detailed or different classifications and associated matters would be dealt with by other members of the family. The potential for different members of the 'family' in the medico-social and multi-dimensional assessment in relation not only to health but also to activities of daily living, as well as the social and physical environment, was recognised. It was demonstrated that effective information could be obtained through use of the ICD and the International Classification of Impairments, Disabilities, and Handicaps (ICIDH) (18), and through use of the codes from Chapter XXI of the Tenth Revision.

The main innovation in the Tenth Revision was the use of an alphanumeric coding scheme of one letter followed by three numbers at the four-character level. This had the effect of more than doubling the size of the coding frame in comparison with the Ninth Revision and enabled the vast majority of chapters to be assigned a unique letter or group of letters, each capable of providing 100 three-character categories. Of the 26 available letters, 25 had been used, the letter U being left vacant for future additions and changes, and for possible interim classifications to solve difficulties arising at the national and international level between revisions.

Another important innovation was the creation towards the end of certain chapters of categories for postprocedural disorders. These identified important conditions that



constituted a medical care problem in their own right. Postprocedural conditions that were not specific to a particular body system continued to be classified in the chapter on 'Injury, poisoning and certain other consequences of external causes'. The Revision included definitions, standards, and reporting requirements related to maternal mortality and to fetal, perinatal, neonatal and infant mortality. It was published in three volumes: one containing the Tabular List, a second containing all related definitions, standards, rules and instructions, and a third containing the Alphabetical Index.

The Tenth Revision Conference discussed the difficulties experienced during the extended period of use of the Ninth Revision, related to the emergence of new diseases and the lack of an updating mechanism to accommodate them. It recognised that it would not be feasible to hold revision conferences more frequently than every 10 years. It also recognised that any changes introduced during the lifetime of the Tenth Revision would need to be considered carefully in relation to their impact on analyses and trends.

### 3.13.10 The WHO Family of International Classifications

Although the ICD is suitable for many different applications, it does not serve all the needs of its various users. It does not provide sufficient detail for some specialties and sometimes information on different attributes of health conditions may be needed. Also, the ICD is not useful to describe functioning and disability as aspects of health and does not include a full array of health interventions or reasons for encounter. Foundations laid by the International Conference on ICD-10 in 1989 provided the basis for the development of a 'family' of health classifications. This was given added momentum during the 1990s by the development of the International Classification of Functioning, Disability and Health (ICF) (19), approved by the World Health Assembly in 2001.

In 2001, the WHO Family of International Classifications (WHO-FIC) was created. At the core of the Family are its reference classifications, currently the ICD and the ICF; the International Classification of Health Interventions (ICHI), now under development, is the third reference classification. The WHO-FIC also includes derived classifications, which provide additional detail to core classifications or are rearrangements or aggregations of terms in core classifications; the WHO has licensed several countries to develop national modifications of the ICD as derived classifications. As well, the WHO-FIC includes related classifications to cover health functions which are not (or are only partially) covered by other WHO-FIC members. The WHO-FIC is supported by a network of Collaborating Centres, based on the former Collaborating Centres for the ICD and the ICF, but continuously expanded by the addition of new centres.

**Table 2:** *Evolution of ICD*



| <b>Iteration</b> | <b>Year</b> | <b>Document</b>                                                                  | <b>Note</b>                                                                                             |
|------------------|-------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 0                | 1891        | Bertillon Classification of Causes of Death                                      | Drafted by International Statistical Institute                                                          |
| 1                | 1900        | Bertillon/International List of Causes of Death                                  | First International Conference for Revision of List of Causes of Death                                  |
| 2                | 1910        | International List of Causes of Death                                            | 179 titles, call for revision every 10 years                                                            |
| 3                | 1920        | International List of Causes of Death                                            |                                                                                                         |
| 4                | 1929        | International List of Causes of Death                                            | Drafted jointly by International Statistical Institute and Health organisation of the League of Nations |
| 5                | 1938        | International List of Causes of Death                                            | 200 titles, additions to infectious and parasitic                                                       |
| 6                | 1948        | International Classification of Diseases, Injuries, and Causes of Death          | Recognition of classification of disease and injury, in addition to causes of death                     |
| 7                | 1955        | International Classification of Diseases, Injuries, and Causes of Death          |                                                                                                         |
| 8                | 1965        | International Classification of Diseases, Injuries, and Causes of Death          |                                                                                                         |
| 9                | 1976        | International Classification of Diseases, Injuries, and Causes of Death          | Trial of supplement on Impairments and Handicaps, and Procedures in Medicine                            |
| 10               | 1990        | International Statistical Classification of Diseases and Related Health Problems | Introduction of alpha-numeric coding scheme, postprocedural disorders; regular interim updates          |
| 11               | 2019        | International Statistical Classification of Diseases and Related Health Problems | Postcoordination (cluster coding); addition of Traditional Medicine, adverse events coding              |

### 3.13.11 Updating of ICD between revisions

As foreshadowed at the Tenth Revision conference, updating of the tenth revision of ICD commenced in 2000. Updating proposals came from, and were carefully considered by, the WHO and Collaborating Centres, including the impact on trends. The updating process has allowed an extended life for the Tenth Revision while maintaining its clinical and scientific currency.

### 3.13.12 Major Steps in the ICD-11 Revision

The revision of ICD-11 has taken place in several phases. First; a list of issues that were known from the use of ICD-10 and that could not be solved in its classification structure was

compiled and possible solutions were formulated. Second; input was received from many scientific groups in the key subject areas, with a focus on clinical perspectives. Finally, centralised editing occurred, aimed at adjusting imbalances in content generated by multiple, independently operating expert groups in the previous phase of the revision, and at ensuring consistency and practicability of the overall structure for mortality and morbidity statistics.

The final version also received input from field testing, Member State comments, and ongoing submissions and processing of proposals.

### 3.13.13 Preparations for the Eleventh Revision

By 2003, it was becoming clear to the WHO and the Collaborating Centres that a further revision of the ICD could no longer be delayed. The extent to which ICD updating could encapsulate emerging developments was limited by the structure of ICD–10, and some issues needed extended development and discussion with expert groups. A special meeting of Collaborating Centres in Helsinki in 2004 discussed the need for a revision and issues to be addressed as part of the revision process. The 2004 WHO-FIC meeting subsequently adopted a revision process work-plan which was progressively developed at ensuing meetings.

In 2007, the WHO formally launched the revision process. Oversight has been provided through a broad-based Revision Steering Group. Technical work has been undertaken by a series of Topic Advisory Groups, with cross-cutting groups examining mortality, morbidity and quality and safety issues. For the first time, a chapter on description of diseases and patterns of diseases from a traditional medicine standpoint has been included.

A Content Model, including a range of components for each ICD entity has been developed, giving a rich foundation for the ICD. Other classifications and terminologies are linked or included where possible to ensure ICD is aligned with them, and items used in other members of the WHO Family of Classifications have been aligned wherever possible. The more traditional statistical classification for mortality and morbidity is obtained from the Foundation Component of ICD–11 as a tabular list. Extension codes are used to limit content volume but still allow detailed classification of disease entities. Supplementary chapters and sections allow capturing on an optional basis information about traditional medicine diagnoses and functioning. Based on the experiences with ICD-9 and ICD-10 an updating mechanism was designed, that allows improvements in user guidance and scientific updates without compromising the statistical use of the classification.

### 3.13.14 References for history of ICD

1. Knibbs G.H. The International Classification of Disease and Causes of Death and its revision. Medical journal of Australia, 1929, 1:2-12.
2. Moriyama IM, Loy RM, Robb-Smith AHT. History of the statistical classification of diseases and causes of death. Rosenberg HM, Hoyert DL, eds. Hyattsville, MD: National Center for Health Statistics. 2011.
3. Greenwood M. Medical statistics from Graunt to Farr. Cambridge, Cambridge University Press, 1948.
4. First annual report. London, Registrar General of England and Wales, 1839, p. 99.
5. Bertillon J. Classification of the causes of death (abstract). In: Transactions of the 15th International Congress on Hygiene Demography. Washington, 1912.
6. International list of causes of death. The Hague, International Statistical Institute, 1940.
7. Roesle E. Essai d'une statistique comparative de la morbidité devant servir à établir les listes spéciales des causes de morbidité. Geneva, League of Nations Health organisation, 1928 (document C.H. 730)
8. Medical Research Council, Committee on Hospital Morbidity Statistics. A provisional classification of diseases and injuries for use in compiling morbidity statistics. London, Her Majesty's Stationery Office, 1944 (Special Report Series No. 248).
9. US Public Health Service, Division of Public Health Methods. Manual for coding causes of illness according to a diagnosis code for tabulating morbidity statistics. Washington, Government Publishing Office, 1944 (Miscellaneous Publication No. 32).
10. Sixteenth annual report. London, Registrar General of England and Wales, 1856, App. p.73.
11. Official Records of the World Health Organisation, 1948, 11, 23.
12. Official Records of the World Health Organisation, 1948, 2, 110.
13. Manual of the international statistical classification of diseases, injuries, and causes of death. Sixth revision. Geneva, World Health Organisation, 1949.
14. Report of the International Conference for the Seventh Revision of the International Lists of Diseases and Causes of Death. Geneva, World Health Organisation, 1955 (unpublished document WHO/HA/7 Rev. Conf./17 Rev. 1.
15. Third Report of the Expert Committee on Health Statistics. Geneva, World Health organisation, 1952 (WHO Technical Report Series, No. 53).
16. Report of the International Conference for the Eighth Revision of the International Classification of Diseases. Geneva, World Health Organisation, 1965 (unpublished document WHO/ICD9/74.4.
17. Manual of the international statistical classification of diseases, injuries, and causes of death, Volume 1. Geneva, World Health Organisation, 1977.
18. International Classification of Impairments, Disabilities, and Handicaps. Geneva, World Health Organisation, 1980.
19. International Classification of Functioning, Disability and Health. Geneva, World Health organisation, 2001
20. Eleventh revision of the International Classification of Diseases, Report by the Director-General, 2019,  
[https://apps.who.int/gb/ebwha/pdf\\_files/WHA72/A72\\_29Add1-en.pdf](https://apps.who.int/gb/ebwha/pdf_files/WHA72/A72_29Add1-en.pdf)

## 3.14 Annex C: Annexes for Mortality Coding

### 3.14.1 International form of medical certificate of cause of death

|                                                                                                                                                                                                     |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------|--------------------------------------------------------------------------------|-------------------------------|---|---|-------------------------------------------------------------------------------------------|----------------------------------|----------------------------------------------------|---|---|---|---|---|---|---|
| <b>Administrative Data</b> (can be further specified by country)                                                                                                                                    |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| Sex                                                                                                                                                                                                 | <input type="checkbox"/> Female                                                                                                                                                                                                                             |                                                  |                |                                                                                | <input type="checkbox"/> Male |   |   |                                                                                           | <input type="checkbox"/> Unknown |                                                    |   |   |   |   |   |   |   |
| Date of birth                                                                                                                                                                                       | D                                                                                                                                                                                                                                                           | D                                                | M              | M                                                                              | Y                             | Y | Y | Y                                                                                         | Date of death                    | D                                                  | D | M | M | Y | Y | Y | Y |
| <b>Frame A: Medical data: Part 1 and 2</b>                                                                                                                                                          |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| <b>1</b><br>Report disease or condition directly leading to death on line a<br><br>Report chain of events in due to order (if applicable)<br><br>State the underlying cause on the lowest used line | <br><br> | a                                                | Cause of death | Time interval from onset to death                                              |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
|                                                                                                                                                                                                     |                                                                                                                                                                                                                                                             | b                                                | Due to:        |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
|                                                                                                                                                                                                     |                                                                                                                                                                                                                                                             | c                                                | Due to:        |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
|                                                                                                                                                                                                     |                                                                                                                                                                                                                                                             | d                                                | Due to:        |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| <b>2</b> Other significant conditions contributing to death (time intervals can be included in brackets after the condition)                                                                        |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| <b>Frame B: Other medical data</b>                                                                                                                                                                  |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| Was <b>surgery</b> performed within the last 4 weeks? <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> Unknown                                                     |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| If yes please specify date of surgery D D M M Y Y Y Y                                                                                                                                               |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| If yes please specify reason for surgery (disease or condition)                                                                                                                                     |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| Was an autopsy requested? <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> Unknown                                                                                 |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| If yes were the findings used in the certification? <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> Unknown                                                       |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| <b>Manner of death:</b>                                                                                                                                                                             |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| <input type="checkbox"/> Disease                                                                                                                                                                    |                                                                                                                                                                                                                                                             |                                                  |                | <input type="checkbox"/> Assault                                               |                               |   |   | <input type="checkbox"/> Could not be determined                                          |                                  |                                                    |   |   |   |   |   |   |   |
| <input type="checkbox"/> Accident                                                                                                                                                                   |                                                                                                                                                                                                                                                             |                                                  |                | <input type="checkbox"/> Legal intervention                                    |                               |   |   | <input type="checkbox"/> Pending investigation                                            |                                  |                                                    |   |   |   |   |   |   |   |
| <input type="checkbox"/> Intentional self harm                                                                                                                                                      |                                                                                                                                                                                                                                                             |                                                  |                | <input type="checkbox"/> War                                                   |                               |   |   | <input type="checkbox"/> Unknown                                                          |                                  |                                                    |   |   |   |   |   |   |   |
| If external cause or poisoning:                                                                                                                                                                     |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   | Date of injury                                                                            |                                  | D                                                  | D | M | M | Y | Y | Y | Y |
| Please describe how external cause occurred (If poisoning please specify poisoning agent)                                                                                                           |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| <b>Place of occurrence of the external cause:</b>                                                                                                                                                   |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| <input type="checkbox"/> At home                                                                                                                                                                    |                                                                                                                                                                                                                                                             | <input type="checkbox"/> Residential institution |                | <input type="checkbox"/> School, other institution, public administrative area |                               |   |   |                                                                                           |                                  | <input type="checkbox"/> Sports and athletics area |   |   |   |   |   |   |   |
| <input type="checkbox"/> Street and highway                                                                                                                                                         |                                                                                                                                                                                                                                                             | <input type="checkbox"/> Trade and service area  |                | <input type="checkbox"/> Industrial and construction area                      |                               |   |   |                                                                                           |                                  | <input type="checkbox"/> Farm                      |   |   |   |   |   |   |   |
| <input type="checkbox"/> Other place (please specify):                                                                                                                                              |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  | <input type="checkbox"/> Unknown                   |   |   |   |   |   |   |   |
| <b>Fetal or infant Death</b>                                                                                                                                                                        |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| Multiple pregnancy                                                                                                                                                                                  |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   | <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> Unknown |                                  |                                                    |   |   |   |   |   |   |   |
| Stillborn?                                                                                                                                                                                          |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   | <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> Unknown |                                  |                                                    |   |   |   |   |   |   |   |
| If death within 24h specify number of hours survived                                                                                                                                                |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   | Birth weight (in grams)                                                                   |                                  |                                                    |   |   |   |   |   |   |   |
| Number of completed weeks of pregnancy                                                                                                                                                              |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   | Age of mother (years)                                                                     |                                  |                                                    |   |   |   |   |   |   |   |
| If death was perinatal, please state conditions of mother that affected the fetus and newborn                                                                                                       |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| <b>For women, was the deceased pregnant?</b> <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> Unknown                                                              |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   |                                                                                           |                                  |                                                    |   |   |   |   |   |   |   |
| <input type="checkbox"/> At time of death                                                                                                                                                           |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   | <input type="checkbox"/> Within 42 days before the death                                  |                                  |                                                    |   |   |   |   |   |   |   |
| <input type="checkbox"/> Between 43 days up to 1 year before death                                                                                                                                  |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   | <input type="checkbox"/> Unknown                                                          |                                  |                                                    |   |   |   |   |   |   |   |
| Did the pregnancy contribute to the death?                                                                                                                                                          |                                                                                                                                                                                                                                                             |                                                  |                |                                                                                |                               |   |   | <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> Unknown |                                  |                                                    |   |   |   |   |   |   |   |

*The International Form of Medical Certificate of Cause of Death*

## The International Form of Medical Certificate of Cause of Death

Additional data that might be necessary for the reporting system of countries can be added to the certificate. The form has a Frame A that serves to report the cause(s) of death, the sequence of causes, the duration of diseases until death, and other conditions contributing to death. Causes of death should be reported with the best available detail. For example, 'birth depression' for a newborn child that died, should be complemented by the reason for the birth depression, as intrapartum asphyxia, or prepartum hypoxaemia.

The Frame B is designed to help to report detail that is relevant to coding and epidemiology analyses for deaths due to external causes, maternal deaths, perinatal deaths, and deaths due to postprocedural conditions. It complements the information in Frame A. The complete reporting of the causes of death is based on an accurate examination of the dead body, the assessment of local circumstances and insight available from health records. Correct establishment of a cause of death and completion of the death certificate requires training that should start at medical school and be refreshed in regular education programmes. Also important is the practical experience in death certification that is gained under the supervision of more experienced colleagues. It is noted that medical certifiers that establish the cause of death may not always be available. Certification by non-physicians may result in a changed pattern of the reported causes of death.

Where the dead body is no longer available for examination, for example due to low coverage of medical staff or traditional rapid burial procedures, a verbal autopsy may provide some limited information on the cause of death. In such a case, a sequence of causes that led to death will rarely be identified, and causes identified with verbal autopsy should be reported separately to those available from formal death certification process.

### 3.14.2 Quick reference guide for the International form of medical certificate of cause of death (MCCD flyer)

Cause of death information serves

- epidemiology and prevention
- managing health care
- comparing health in different populations

Certification of death is one of the first steps in getting an overview of the health of people. The diseases or conditions recorded on a death certificate represent the best medical opinion. A properly completed cause-of-death certificate provides a description of the order, type and association of events that have resulted in the death. The diagnoses reported on the certificate are coded with the International Classification of Diseases. The filling in of the cause-of-death certificate is independent of the revision of ICD. This coded data is analyzed and used both nationally and internationally no matter what language was used to complete the certification.

#### **Cause of Death on the certificate - how to fill in?**

**Frame A:** Death certificates may look different in most countries. But the section on the cause of death is identical worldwide. Frame A has two parts, called Part 1 and Part 2, and a section to record the time interval between the onset of each condition and the date of death.

**Part 1** - is used for diseases or conditions that form part of the **sequence of events** leading directly to death. The immediate (direct) cause of death is entered on the first line, 1(a). There must always be an entry on line 1(a). The entry on line 1(a) may be the only condition reported in Part I of the certificate. Where there are **two or more conditions** that form part of the sequence of events leading directly to death, each event in the sequence should be recorded on a separate line.

In any case you must record the disease, injury or external cause that resulted in the death. Do not record only the **mode of dying**, such as cardiac arrest, respiratory failure or heart failure.

Try to be as specific as you can. **“Unknown”** cause of death should be recorded in cases where thorough testing or autopsy examination the cause of death cannot be determined. Stating “Unknown” is better than any speculation on the possible cause of death.

Always fully spell out all terms. **Abbreviations** can be interpreted in different ways. Terms such as “suspected” or “possible” are ignored in evaluation of the entries. For example “suspected Diabetes” will be interpreted as “Diabetes”. The four lines may not provide enough space for the chain of events. Do not waste space with **unnecessary words**. Some clinical terms are very vague. For example, “tumour” does not specify behaviour (see also last page of this flyer).

**Duration** - is the time interval between the onset of each condition that is entered on the certificate (not the time of diagnosis of the condition), and the date of death. The duration information is useful in coding certain diseases and also provides a useful check on the order of the reported sequence of conditions.

**Part 2** - is used for conditions that do not belong in Part 1 but their presence contributed to death.

**Frame B:** Some detail is frequently forgotten in Part 1 and 2 (Frame A). Separate detailed questions ask for detail such as previous surgery, mode of death or place of occurrence. Frame B is not shown in this information sheet. It is self-explanatory.

### **Cause of Death on the certificate - step by step**

- **Start** at line 1(a), with the immediate (direct) cause, then go back in time to preceding conditions until you get to the one that started the sequence of events. You will get very close to the time the patient was healthy.
- **Now**, you should have reported the underlying or originating cause on the lowest used line and a sequence of events leads from the underlying cause up to the immediate (direct) cause in the first line 1(a).
- **Finally**, record the time interval between the onset of each condition entered on the certificate and the date of death. Where the time or date of onset is not known you

should record a best estimate. Enter the unit of time (minutes, hours, days, weeks, months, years).

## Example

| Frame A: Medical data: Part 1 and 2                                                                                                                                                                 |          |                                    |                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------|------------------------------------------|
| <b>1</b><br>Report disease or condition directly leading to death on line a<br><br>Report chain of events in due to order (if applicable)<br><br>State the underlying cause on the lowest used line |          | <b>Cause of death</b>              | <b>Time interval from onset to death</b> |
|                                                                                                                                                                                                     | <b>a</b> | Cerebral haemorrhage               | 4 hours                                  |
|                                                                                                                                                                                                     | <b>b</b> | Due to:<br>Metastasis to the brain | 4 months                                 |
|                                                                                                                                                                                                     | <b>c</b> | Due to:<br>Breast cancer           | 5 years                                  |
|                                                                                                                                                                                                     | <b>d</b> | Due to:                            |                                          |
| <b>2</b> Other significant conditions contributing to death (time intervals can be included in brackets after the condition)                                                                        |          |                                    |                                          |

### Example

- **Write clearly** and do not use abbreviations.
- Be sure the information is **complete**.
- **Do not speculate** on the cause of death.
- Do not fill in laboratory results or statements like “found by partner” (there may be separate fields on the form for this kind of information).
- One condition per line should be sufficient.

### Frequently used ill-defined terms



| <b>Term</b>                    | <b>Instruction</b>                                                                                                                                                                |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Accident</b>                | Specify <b>circumstances</b> . Specify <b>intent</b> , as car accident, suicidal, or assault. Specify <b>place</b> of occurrence.                                                 |
| <b>Alcohol, drugs</b>          | Specify <b>use</b> : long term or single, addiction                                                                                                                               |
| <b>Complication of surgery</b> | Specify <b>disease</b> that was the reason for surgery                                                                                                                            |
| <b>Dementia</b>                | Specify <b>cause</b> : e.g. due to Alzheimer disease, cerebrovascular, Lewy body                                                                                                  |
| <b>Hepatitis</b>               | Specify <b>course, etiology</b> : acute or chronic, alcoholic. If viral (specify type A, B, C..)                                                                                  |
| <b>Infarction</b>              | Specify <b>site</b> : heart, brain... Specify <b>cause</b> : arteriosclerotic, thrombotic, embolic                                                                                |
| <b>Infection</b>               | Specify: primary or secondary, causative <b>organism</b> . If <b>primary</b> : specify bacterial, viral, fungal or parasitic. If <b>secondary</b> : specify the primary infection |
| <b>Leukaemia</b>               | Specify: <b>type</b> e.g. myeloid, monocytic, lymphoid, also whether acute or chronic                                                                                             |
| <b>Pneumonia</b>               | Specify: primary, aspiration, <b>cause</b> , causative <b>organism</b> . If due to immobility: specify the cause of immobility                                                    |
| <b>Pulmonary embolism</b>      | Specify <b>cause</b> of embolism. If <b>postsurgical</b> or due to <b>immobility</b> , indicate the disease that was a cause for surgery or of immobility                         |
| <b>Renal failure</b>           | Specify: acute, chronic or terminal, underlying <b>cause</b> of insufficiency, such as arteriosclerosis or infection. If due to immobility: specify the cause of immobility       |
| <b>Thrombosis</b>              | Specify: arterial or venous. Specify: the blood vessel. If postsurgical or immobility: specify disease that was a cause for surgery or immobility                                 |
| <b>Tumour</b>                  | Specify: behaviour, location, metastases                                                                                                                                          |
| <b>Urinary tract infection</b> | Specify: site in the urinary tract, causative <b>organism</b> , underlying <b>cause</b> of infection. If due to immobility: specify the cause of immobility                       |

### 3.14.3 Suggested additional details of perinatal deaths

|                                             |  |  |  |  |  |                                        |   |   |   |   |   |                                    |       |
|---------------------------------------------|--|--|--|--|--|----------------------------------------|---|---|---|---|---|------------------------------------|-------|
| <b>Identifying particulars</b>              |  |  |  |  |  |                                        |   |   |   |   |   |                                    |       |
| Child was born live on                      |  |  |  |  |  | D                                      | D | M | M | Y | Y | at hh:mm                           | hours |
| Child was stillborn on                      |  |  |  |  |  | D                                      | D | M | M | Y | Y | at hh:mm                           | hours |
| <input type="checkbox"/> Died before labour |  |  |  |  |  | <input type="checkbox"/> During labour |   |   |   |   |   | <input type="checkbox"/> Not known |       |

|               |  |  |   |   |   |   |   |   |  |  |  |  |
|---------------|--|--|---|---|---|---|---|---|--|--|--|--|
| <b>Mother</b> |  |  |   |   |   |   |   |   |  |  |  |  |
| Date of birth |  |  | D | D | M | M | Y | Y |  |  |  |  |

|                                                        |  |  |  |  |  |                                                                                                                                                              |  |  |   |   |   |   |   |   |
|--------------------------------------------------------|--|--|--|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|---|---|---|---|---|---|
| <b>Number of previous pregnancies</b>                  |  |  |  |  |  | <b>Date of last pregnancy</b>                                                                                                                                |  |  | D | D | M | M | Y | Y |
| Live birth _____<br>Stillbirth _____<br>Abortion _____ |  |  |  |  |  | <b>Outcome of last previous pregnancy</b><br><input type="checkbox"/> Live birth<br><input type="checkbox"/> Stillbirth<br><input type="checkbox"/> Abortion |  |  |   |   |   |   |   |   |

|                                         |  |  |  |  |  |   |   |   |   |   |   |  |  |
|-----------------------------------------|--|--|--|--|--|---|---|---|---|---|---|--|--|
| <b>1st day of last menstrual period</b> |  |  |  |  |  | D | D | M | M | Y | Y |  |  |
|-----------------------------------------|--|--|--|--|--|---|---|---|---|---|---|--|--|

|                                                                                                      |  |  |  |  |  |                                                                                                   |  |  |  |  |  |  |  |
|------------------------------------------------------------------------------------------------------|--|--|--|--|--|---------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| <b>Delivery</b>                                                                                      |  |  |  |  |  | <b>Antenatal care, two or more visits</b>                                                         |  |  |  |  |  |  |  |
| <input type="checkbox"/> Normal spontaneous vertex<br><input type="checkbox"/> Other (specify) _____ |  |  |  |  |  | <input type="checkbox"/> Yes<br><input type="checkbox"/> No<br><input type="checkbox"/> Not known |  |  |  |  |  |  |  |

|                                          |  |  |  |  |  |                                                               |  |  |  |  |  |  |
|------------------------------------------|--|--|--|--|--|---------------------------------------------------------------|--|--|--|--|--|--|
| <b>Attendant at birth</b>                |  |  |  |  |  |                                                               |  |  |  |  |  |  |
| <input type="checkbox"/> Physician       |  |  |  |  |  | <input type="checkbox"/> Other trained person (specify) _____ |  |  |  |  |  |  |
| <input type="checkbox"/> Trained midwife |  |  |  |  |  | <input type="checkbox"/> Other (specify) _____                |  |  |  |  |  |  |

|                                       |  |  |  |  |  |                                                         |  |  |  |  |  |  |
|---------------------------------------|--|--|--|--|--|---------------------------------------------------------|--|--|--|--|--|--|
| <b>Child</b>                          |  |  |  |  |  |                                                         |  |  |  |  |  |  |
| <input type="checkbox"/> Single birth |  |  |  |  |  | <input type="checkbox"/> Second twin                    |  |  |  |  |  |  |
| <input type="checkbox"/> First twin   |  |  |  |  |  | <input type="checkbox"/> Other multiple (specify) _____ |  |  |  |  |  |  |

*Suggested Additional Details Perinatal Deaths*